

Visualizing Exercise Data from Combat Exergame for Exploring the Insight from Personal Informatics with Large Language Models

Zong-Ying Li

Department of Computer Science and
Information Engineering
National Taipei University of
Technology
Taipei, Taiwan
wuxinde409.co@gmail.com

Yen-Hua Lai

Department of Interaction Design
National Taipei University of
Technology
Taipei, Taiwan
lyhhowardlai@gmail.com

Chi-Yu Lin

Department of Interaction Design
National Taipei University of
Technology
Taipei, Taiwan
linchiyu76@gmail.com

Chien-Hsing Chou

Department of Mathematics and
Information Education
National Taipei University of
Education
Taipei City, Taiwan
chchou@mail.ntue.edu.tw

Ping-Hsuan Han

Department of Interaction Design
National Taipei University of
Technology
Taipei, Taiwan
han@ntut.edu.tw

Abstract

Previous research has explored the potential of personal informatics and data visualization to enhance reflection. However, these studies primarily focus on improving user performance in traditional sports. In this work, we present a visualization system for innovative sports integrating sports data collected from a VR combat exergame with personalized narratives generated by Large Language Models (LLMs). We conducted a user study with Data Visualization and LLM Suggestions to evaluate their impact on self-reflection and reflection. The results show that our system facilitated data exploration and comparison with reflection. In addition, this study reveals the potential of combining LLMs and VR-based data visualization to provide personalized feedback for users.

CCS Concepts

• **Human-centered computing** → **Heuristic evaluations.**

Keywords

personal informatics, reflection, data visualization, large language model, Exergame

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1 Introduction

In the combat sports environment, additional sensors and cameras are required to collect the data to perform visualization. Worsey et al. [28] placed 9DOF inertial sensors under boxing gloves to capture data on thrown punches, while the other method of capturing punching data was to use piezoresistive pressure sensors to measure the impact of punches, and inertial sensors were used to track the trajectory of punches [18]. However, this approach, which involves additional equipment, increases the overall cost of the system and requires more time for hardware configuration, thereby reducing the opportunities to collect raw data. On the other hand, having access to raw data allows us to observe more detailed aspects of motion, providing deeper insights into performance and enabling more precise analysis. Although data visualization technology has become highly advanced and widespread, the recent advancements in VR have also sparked exploration into integrating and analyzing exergame data to enhance exercise motivation and boost sports participation [3][5], and in order to efficiently summarize complex data, large language models (LLMs) offer integrated textual explanations, providing users with additional information and insights [31].

In recent years, the rapid development of artificial intelligence, particularly LLMs, has established them as one of the most popular and widely adopted technologies. However, there has been relatively little research on the application of LLMs in the formative of innovative sports data, especially in utilizing personal data to enhance self-reflection behaviors. Existing studies predominantly rely on publicly available datasets or annual summaries of team sports data. Sarfati et al [21] extracted key events from Premier League matches and used GPT-3 to generate sports highlights narratives, Caron et al [4] utilized sports data from StatsBomb with

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GPT-NeoX-20B to simulate players' movements and match scenarios. Strömel et al. [24] employed three different data representation methods, text, chart, and a combination of text and chart, to demonstrate that using LLM and visualizations is most effective in eliciting reflection. Meanwhile, these studies mainly focus on integrating traditional sports data with LLMs. Recently, the development of virtual reality (VR) and Esports has brought new trends for innovative sports that already include the personal informatics, creating the possibility for more personalized and effective approaches in sports. Langer et al. [14] cooperate with the training of the reaction ability required for combat sports in virtual reality. Xu et al. [30] introduced VirusBoxing as a form of innovative technology sports, enabling participants to engage in VR-based High-Intensity Interval Training (HIIT) while providing an engaging and enjoyable exercise experience. In this work, we collected personalized data by combining VR and the combat exergame and analyzed it using LLM. We analyzed the collected data, both formative and summary data, using the prompts. This process enabled the generation of specialized, personalized recommendations based on the formative data.

2 Related Works

2.1 Data Visualization for Traditional Sports

In the field of sports, data serves as a critical resource for enhancing athletic performance. Legg et al. [15] analyzes data from rugby matches and presents glyph designs to represent events and behaviors during the games. The system utilizes symbolic glyphs to encode multivariate data, seamlessly combining graphical representation with intuitive comprehension. This design approach enables users to interpret complex datasets with enhanced clarity rapidly. This method utilizes the tracking data of players and the ball during a soccer game to visualize team formations and player movements. The visualization systems provided a method that coaches and players could efficiently evaluate team general performance and make strategic decisions with greater precision[2]. With technological advances, one of the primary challenges is to effectively visualize these data in order to create actionable insights for coaches and athletes has emerged as a prominent focus of contemporary research. Most studies have used competitive sports as the primary focus for data visualization. As highlighted by [9], the core visualization tasks in competitive sports have been analyzed, and key visualization techniques have been systematically summarized. Despite the introduction of numerous sports visualization systems and objectives, significant gaps remain in addressing the unique requirements of certain sports. This paper aims to explore the visualization of data specifically within the domain of combat sports.

The collection and analysis of Personal Informatics (PI) in sports data play a crucial role in facilitating self-awareness and behavioral improvements. Rapp et al. [20] demonstrated that elite athletes can integrate their prior knowledge with personal informatics data to enhance their performance, while amateur athletes can use simple PI systems to gain a basic understanding of their abilities. Pilegg et al. [19] explore ice hockey by generating Shot Maps to identify optimal shooting positions and assist in game strategy formulation. Bialkowski et al. [2] combine player positions to create formation maps, allowing teams to analyze opponents' formations and tactical

adjustments during matches. Several studies have investigated data analysis in traditional team sports, focusing on gaining overall insights into team strategies. In personal sports, particularly combat sports like boxing, judo, and taekwondo, individual performance plays a critical role. However, the complexity and cost of collecting personal data remain significant challenges. These sports studies often rely on additional hardware devices to capture motion data, such as sensors and piezoresistive pressure sensors. Menzel et al. [18] employ piezoresistive pressure sensors to measure comprehensive motion data, including punch force, and employ a Kistler Force Plate to evaluate accuracy and practicality when integrating different hardware. Emad et al. [10] utilized Kinect v2 to collect motion data and provide trainees with real-time movement feedback, enhancing the learning efficiency of karate techniques. Thanh et al. [26] obtained coordinates of the joints from the Kinect camera to assist participants in receiving movement feedback for Vietnamese traditional martial arts and provided performance evaluation through the system. Wu et al. [29] proposed a system that uses an RGB camera to capture athletes' motion images, applies deep learning for 3D pose estimation, and employs recurrent neural networks for pose forecasting to predict future movements. Finally, the system utilizes mixed reality technology to visualize the motions, assisting martial arts practitioners in precise skill acquisition. According to these studies, previous research on personal informatics in sports has primarily focused on traditional sports, using sensor devices to collect movement data for performance analysis and feedback. In our study, we aim to explore the relationship between personal informatics and a few movement data obtained from innovative technology sports.

2.2 Reflection through Sports Data and LLMs

Previous studies have focused on Virtual Reality (VR) to collect personalized data through Exergame or VR-Based Combat Sports. Langer et al. [14] propose an innovative training method for reaction time in virtual environments, which is an important feature in combat sports. Furthermore, while these data are primarily used for performance analysis and enhancement, less research has explored their potential for fostering self-reflection in sports.

Reflection, as a key concept in personal behavior, has been shown to improve performance and self-recognition in several studies. Tuomainen [27] discovers that self-reflection allowed students to identify their strengths, fostering a positive attitude toward the preparation of subsequent presentations. Tan et al. [25] highlighted how guided self-reflection improved performance in elite athletes. Narrative or storytelling content can further facilitate reflection by helping users contextualize the relationship between data and behavior. Farrow et al. [11] demonstrated how the conversion of personal social media data into positive and negative narratives allowed users to understand emotional changes over a week, enhancing their self-awareness. The application of LLMs in processing sports data has grown significantly in recent years. For sports broadcasting, GPT-3 has been used to generate narratives of highlights from football matches, enhancing the viewing experience by providing engaging and contextual summaries [21]. Caron et al. [4] employed GPT-NeoX-20B to intuitively present complex tactical data, enabling viewers to better understand the movements of the

players on the field. Although these studies highlight the significant contributions of LLMs in processing large open datasets, they also reveal limitations in providing personalized reflection insights and detailed analyses of individual motor states and movements. Regarding improving exercise participation, Geurts et al. [12] apply LLM and GPX sports data to generate personalized motivational messages in a VR exercise environment, in order to reduce loneliness and enhance their exercise motivation.

Innovative Sports refers to a type of sport that requires players to apply the technology equipment. Olympic[8] has incorporated various Esports into official competitions, while [23] actively promotes new forms of sports by integrating technology with human. Schättin et al. [22] presented Sphery Racer, incorporating on-body feedback technology to encourage home exercise participation. Other studies have enhanced immersion through Encountered-Type Haptic Feedback. MovableBag[17] simulates a boxing opponent or coach to improve the training experience. In contrast, Metapunch X[6] integrates exertion interaction and multi-screen spectating to enhance both player engagement and audience experience. Xu et al. [30] combined High-Intensity Interval Training (HIIT) with VR, creating an interactive boxing experience. [7] developed an exergame utilizing sensory substitution technology to enhance the sports experience. Previous studies on Innovative Sports have not only demonstrated their potential to increase exercise motivation but have also explored the role of haptic feedback in improving the overall sports experience.

Prior research has explored how personal data can enhance reflection and the use of LLMs to analyze sports data, primarily in traditional sports. Moreover, some research on innovative sports has tended to enhance exercise motivation and sports experiences. However, there has been less intent on how personal data collected by exergames could facilitate reflection and help users better understand their own data. Inspired by these studies, we explore how the limited personal sport data, specifically head and hand coordinates instantly collected during a VR combat exergame can be transformed into an effective data visualization system and integrated with LLMs to generate textual narratives, helping to facilitate reflection and self-reflection.

3 Design Consideration

The study aims to focus on the different kinds of personal data generated by exergames of combat sports in VR, and through the visualization of those data and the use of LLMs, it aims to generate suggestions for facilitating users' self-reflection to understand themselves. Addressing the challenge of improving data interpretation and user comprehension, the system incorporates interactive interface to enhance engagement with the data. Furthermore, the study evaluates the integration of data visualization and narrative suggestions to ensure that the LLM provides specific, personalized recommendations rather than relying solely on open datasets to generate insights.

4 System Overview

Boxing, a renowned and historic combat sport, is characterized by its strict rules allowing punches only with the hands and targeting

specific areas of the opponent's body. This article leverages user-generated boxing data within a VR environment to analyze boxing styles and provide tailored recommendations for improvement. We created a visualization system based on personal informatics data. We started by allowing users to view two types of data: summary and formative. These were then integrated into the operable system interface that represents the punching style while facilitating reflection through a system. With the appearance of many technological devices and the increasing popularity of wearable devices, the collection of personal data is a common and daily behavior. Still, for the collected data, users generally only understand the existence of these numerical values without actually understanding the deeper information. However, for them to gain a greater understanding of data so they can realize what data can do, data sense, an interactive approach, becomes a suitable goal to explore [16].

4.1 Data Processing

We collected 468 punches from users who completed the exergame and organized them into adequately formatted folders, each assigned with specific time-based labels. Our system allows users to select their data files through a JSON interface.

Summary Data: Provides personal informatics by calculating seven key values, which are displayed as a bar chart in Fig. 1(A). The X-axis represents each sports data value range, from minimum to maximum, while the Y-axis shows the number of users within each range. Percentages displayed at the bottom of each graph indicate outcomes derived from a single comparison across all users' data. This visualization lets users quickly understand their performance level relative to the entire user group.

Formative Data: The analysis charts are based on data collected from the frame numbers of the head-mounted display (HMD). By processing the raw data, we generated three graphs depicting the user's punch speed, power, and reaction time for each punch. These graphs are combined with a scatter chart to visualize the user's movement positions over time (as shown in Fig. 2).

4.2 Personal Informatics Data Visualization

The x-axis is the user's left and right movement distance, the z-axis is the user's forward and backward movement distance, and the y-axis is the user's up and down movement

Punch Animation: This 3D animation is to display all recorded frames in the 3D space through animation. The analysis centers on formative data from the user's hand and head movements, with animations illustrating the punch trajectories. Green nodes represent the left hand, while the right hand is shown by blue nodes, providing details of each motion in Fig. 2. We also developed an additional button and two Spinbox components. The Frame Number spinbox can specify how many frames will appear on the screen at one time to help understand the animation, the All button can display all the frames and let people see all the action paths within the period, and the last one, the speed level spinbox, indicates that by changing the time interval of each picture, interval means Animation interval time, max interval default is 200 milliseconds. Therefore, we set the animation speed to five levels to speed up or slow down the data displayed in the animation.

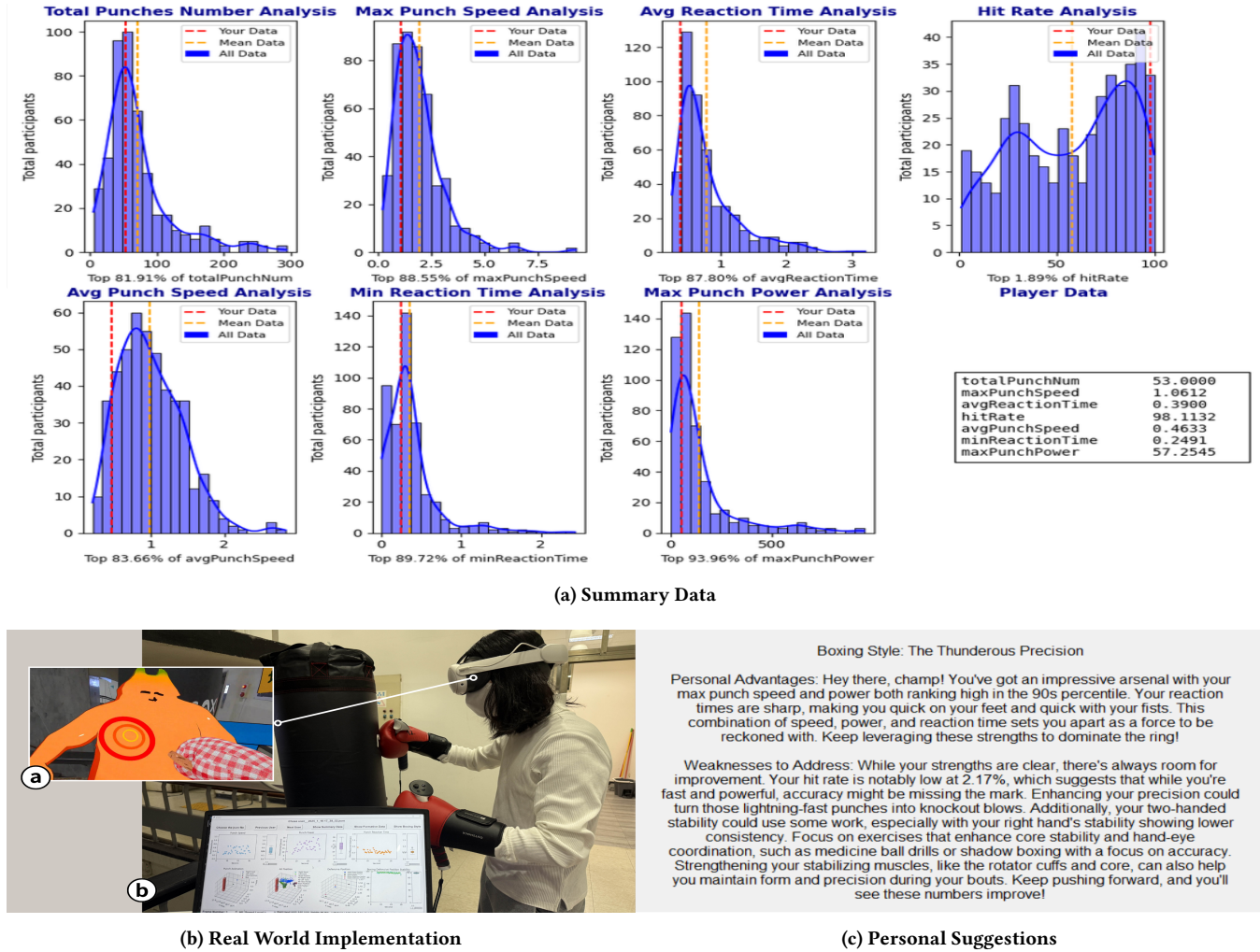


Figure 1: Section A displays the types of values recorded as summary data, Section B shows (a) the exergame interface during operation, followed by (b) the data visualization interface. Section C presents the personalized suggestions generated by the LLM.

All Position: The All position displays all data related to the formative movements. Initially, the positions of the right hand, left hand, and head are based on the user's punches in their actual starting positions. At the same time, displays all tracked movement data of both hands, and the head within 30 seconds, allowing users to review the complete trajectory of their actions. Additionally, the Start and End positions capture the initial and final positions of the head, enabling users to analyze changes in head movement throughout the activity in Fig. 2.

Defensive Position: This section presents the graph representing the user's defensive position following each punch to the bag. To ensure the correct defensive posture, we record the positions of both hands in their defensive state. This is achieved by identifying the point closest to the head and the subsequent point further away from the head in each frame, enabling precise calculation of the hand's defensive position. Finally, we applied the Min-Max

Normalization method to the defensive position coordinates of each punch and calculated the stability of both hands in Fig. 2.

Personal Suggestions: We combine the summary data and the formative data with the prompt (Appendix A) to generate personalized recommendations. These recommendations include three key aspects: Boxing Style, Personal Advantages, and Weaknesses to Address. By providing these insights and feedback, we allow users to understand their data in Fig. 1(C).

5 Preliminary Study

To enhance self-reflection and reflection on personal data, we developed a system with a normal distribution-based visualization interface, incorporating data from multiple individuals for comparison. Our study explores how combat exergame data can be visualized to support reflection. In an experiment, (N=18) participants interacted with the system, which presented their VR exergame data within

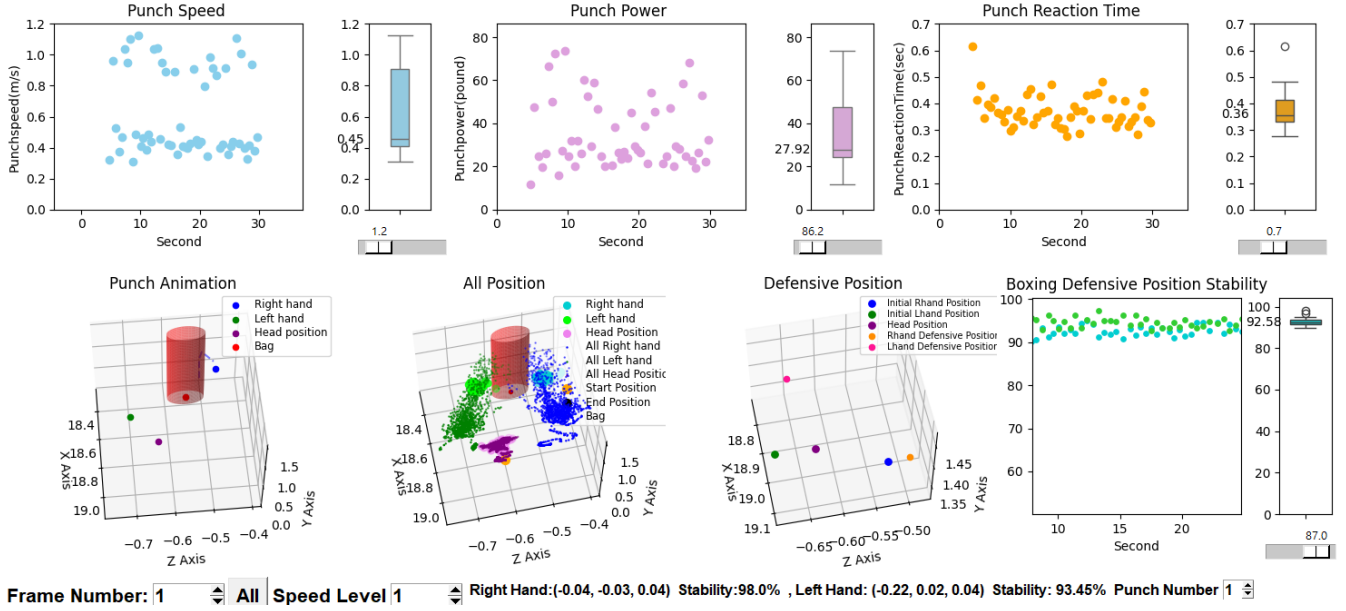


Figure 2: The charts generated using formative data display punch speed, force, and reaction time in the upper section, while the lower section illustrates the movement trajectories of both hands and the defensive position.

the context of a normal distribution derived from ($n=468$) individual data sets. The data was displayed through three interfaces: Base-line (BL) only viewed the VR exergame screen without seeing any data visualization, Data Visualization (DV), and Data Visualization + LLM Suggestions (DV+LLMs), which were evaluated for their effectiveness in fostering self-reflection and data understanding. The results indicate that the Data Visualization + LLM Suggestions interface provided the best support for comprehending personal data.

Participants first filled out a basic questionnaire to confirm whether they had prior VR experience and whether they were VR developers. They were also asked to agree to participate in the experiment and authorize using their personal data. A brief introduction to the VR combat exergame system was provided, which operates with a head-mounted display (HMD) and boxing gloves. Participants were assisted in wearing the HMD, adjusting the straps, and putting on the gloves before stepping in front of the punching bag. The test was divided into three experiments. The first experiment serves as the baseline, where participants experience the combat exergame without accessing any visualization system. The second experiment involves participants experiencing the combat exergame and then reviewing their personal data, as illustrated in Fig. 1(A) and Fig. 2, which showcase different types of data visualizations. In the final experiment, participants review the visualization interface presented in Fig. 1(A) and Fig. 2 after playing the combat exergame and receive additional insights. These include personalized style recommendations generated from summary data and formative data, along with an introduction to their performance strengths and individualized improvement suggestions, as shown in Fig. 1(C), with prompts provided in the appendix A. Hardware implementation We used a computer with intel i5-11400H CPU, RTX

3050 Ti GPU, and 16GB RAM to implement the visualization system. We also use 12oz boxing gloves and a punching bag in combination with Meta Quest 3 and their VR controllers. The Participants first completed the Self-Reflection and Insight Scale (SRIS) [13] using a six-point scale (1 = strongly disagree, 6 = strongly agree) as part of the pre-study evaluation. A counterbalanced approach, based on a Latin-square study design, was then employed to systematically include all arrangements of balanced conditions across the three experiments. After each experiment, participants completed both the SRIS and the Technology-Supported Reflection Inventory (TSRI) [1], using a seven-point scale (1 = strongly disagree, 7 = strongly agree) to evaluate the effectiveness of the systems in facilitating personal reflection.

6 Results and Discussion

This study utilized two different questionnaires, the TSRI and SRIS, to analyze the system and explore how to effectively visualize personal data in combat exergame to promote reflection and enhance self-reflection ability. To further examine the impact of different interfaces on users' reflective thinking, we applied Mauchly's Test of Sphericity to verify the assumptions for repeated measures ANOVA (RM-ANOVA) and then conducted it for analysis. In Fig. 3, we illustrate that we calculated the TSRI scores using Single-factor ANOVA tests. There is a significant difference between using and not using the system (BL: mean = 4.05, SD = 1.47; DV: mean = 5.20, SD = 0.96; DV+LLMs: mean = 5.46, SD = 1.04; $F = 7.287$, $P = 0.002$). This indicates that using the system can promote users' reflection. A closer examination of each subcategory reveals that Exploration has the strongest significance (BL: mean = 4.31, SD = 1.68; DV: mean = 5.63, SD = 1.02; DV+LLMs: mean = 5.93, SD = 0.88; $F = 8.535$, $P < 0.001$), while Insight is weaker (BL: mean = 3.37, SD = 1.50; DV: mean =

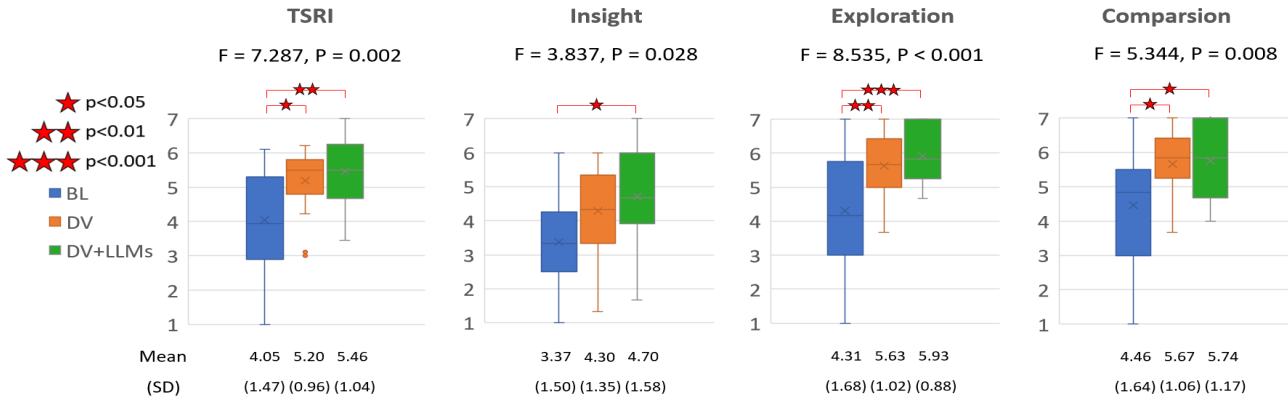


Figure 3: Result of TSRI

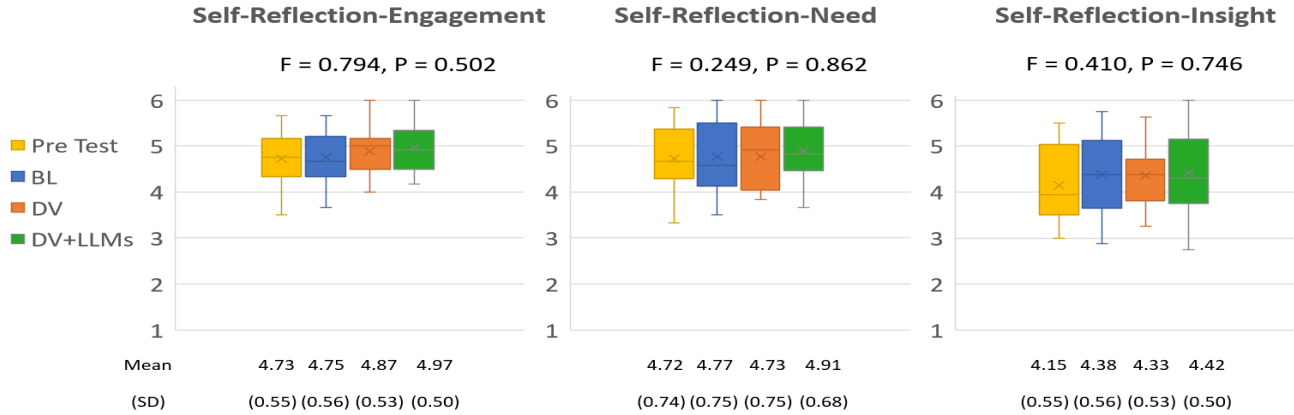


Figure 4: Result of SRIS

4.30, SD = 1.35; DV+LLMs: mean = 4.70, SD = 1.58; $F = 3.837$, $P = 0.028$). The results show that significant differences in Insight only appear when suggestions from the LLM are included. This suggests that combining the system with LLM-generated suggestions helps users gain insights. Additionally, Comparison also shows significant differences (BL: mean = 4.46, SD = 1.64; DV: mean = 5.67, SD = 1.06; DV+LLMs: mean = 5.74, SD = 1.17; $F = 5.344$, $P = 0.008$). Fig. 4 shows that SRIS scores were compared in the experiment using Single-factor ANOVA tests. The results indicate no significant differences in SRIS scores, suggesting that the system does not improve self-reflection. The lack of improved self-reflection ability might be because developing such skills requires a longer-term process. While short-term changes may help users reflect, they are not enough to improve self-reflection ability. This is consistent with the SRIS results, which showed no significant differences when evaluating the system.

We discuss the effects and limitations of the three interfaces mentioned earlier, using the numerical and statistical analyses presented in the results. **Baseline interface** The findings from the SRIS scores

indicate that this interface did not significantly influence participants' self-reflection. A few participants noted that the lack of clear data presentation or intuitive textual descriptions in this interface made it difficult to comprehend, particularly when viewing it individually. Additionally, the low average score for TSRI-I suggests that this interface's straightforward and monotonous design was far less effective at enhancing insight than the other two interfaces. However, the TSRI shows a significant influence on reflection. **Data Visualization interface** The study demonstrated that the visualized interface provided meaningful support for self-reflection. Some participants reported that the Punch Animation, which displayed all movement data for the left hand, right hand, and head within the VR environment, significantly aided their understanding of their performance. Limitations: The calculation of stability in the Defensive Position relied on a min-max normalization method. However, the resulting stability values may not perfectly align with real-world positions, potentially introducing discrepancies in the interpretation of the data. **Data Visualization+LLM Suggestion interface** The addition of LLM-generated suggestions enhanced this interface

by incorporating textual descriptions and percentages to highlight participants' strengths. Some participants suggested that including demographic factors such as age and gender in the suggestions could further improve the contextual relevance and meaning of the feedback. Limitations: The style of prompts provided to the LLM can significantly influence the suggestions it generates. This variability caused some participants to feel that certain strengths were exaggerated, which impacted their perception of the feedback's accuracy. To address these issues in future work, we plan to design a more visually appealing user interface and incorporate a wider variety of data visualization styles. Furthermore, the system is intended for implementation in gyms and sports facilities, allowing users interested in exploring their personal data to benefit from this advanced data visualization and analysis platform.

7 Conclusion and Future Work

In this study we proposed a system that visualizes exercise data collected from a combat exergame and incorporates personalized narratives generated by LLM. To evaluate the impact of the system, we recruited 18 participants to experience three interfaces. The results indicate that DV and DV+LLMs enhance the exploration of personal data. This improvement might be attributed to the visualization on our system, which allows users to more easily reflect on and compare their movements and data. Although the DV+LLMs interface can further generate personalized insights through LLMs, providing participants with new perspectives, the results show no significant difference in the impact on reflection abilities between DV and DV+LLMs. Therefore, future research could explore whether different prompts can generate more detailed and easily understandable sentences, allowing LLM-generated suggestions to cause an additional influence on users.

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A Prompt for Personal Suggestion

This section presents the prompt used to provide personal informatics to GPT-4.

"Based on the following normalized feature data and summary percentage data, please act as a professional boxing coach with expertise in analyzing boxing performance. The Percentage Series represents the percentile rank compared to all other data, indicating the percentage of performance this user has surpassed. The remaining data under formative data reflects key metrics normalized for analytical purposes. Your task is to provide: First, a creative and descriptive name for the boxing style that fits the user's performance metrics. Second, A thoughtful and motivating sentence that encourages the user to reflect on their performance, highlighting strengths and areas for improvement, and helping them quickly

and comprehensively understand their data while feeling a sense of recognition and alignment with the analysis. Third, point out the current numerical weaknesses in not only the summary data in the Percentage Series but also the normalized two-handed stability can be used as part of the weaknesses, to pinpoint which stability sections are susceptible to instability before, during and after the time period. That should be addressed to have a better boxing performance, suggesting reasonable ways to strengthen them, such as which values to improve the basic boxing movements or certain parts of the muscle training that can be very helpful and improve the weaknesses. Title only needs start at Boxing Style: and Personal Advantages: and Weaknesses to Address: and no additional punctuation should be added to the three titles mentioned earlier. Make sure all feedback text sounds a natural easy-going and conversational tone."